IMDB MOVIE REVIEW SENTIMENT ANALYSIS

Natural Language Processing and Machine Learning Project

1. Introduction and Objective

The objective of this project is to develop a machine learning system that classifies movie reviews as either positive or negative using Natural Language Processing (NLP).

The project investigates text preprocessing techniques, feature extraction methods, and different classification algorithms. The models are evaluated using cross-validation and a separate held-out test set to determine their effectiveness in sentiment classification.

The main objectives are:

- Perform exploratory data analysis on movie reviews.
- Evaluate the impact of text preprocessing techniques.
- Compare CountVectorizer and TF-IDF feature representations.
- Train and tune multiple machine learning classifiers.
- Evaluate model performance using standard classification metrics.
- Analyze influential text features and misclassified reviews.

2. Dataset Description and Exploratory Data Analysis

The project uses the IMDB Dataset of 50K Movie Reviews, which contains 50,000 movie reviews labeled as positive or negative.

Dataset characteristics:

- Total reviews: 50,000
- Input feature: Review text
- Target variable: Sentiment
- Positive reviews: 25,000
- Negative reviews: 25,000
- Classification type: Binary classification

Exploratory Data Analysis (EDA) examined the dataset's dimensions, missing values, class distribution, duplicate reviews, review lengths, and examples of review text.

The original dataset contains 418 duplicate review rows. Identical review texts were retained, but the train-test split was constructed to prevent identical review texts from appearing in both sets. This helps reduce the risk of data leakage.

The final dataset split contains:

- **Training set:** 40,000 reviews
- **Testing set:** 10,000 reviews

The original CSV dataset is stored locally and is not included in the repository.

3. Data Preprocessing and Feature Engineering

Text preprocessing was investigated to understand its impact on sentiment classification. The experiments examined HTML tag removal, lowercasing, punctuation removal, and stop-word removal.

HTML tags such as
 were present in the reviews, making HTML stripping a useful preprocessing step.

3.1 Preprocessing Ablation Study

Five-fold cross-validation was used to compare preprocessing configurations.

Configuration	Mean Cross-Validation Accuracy
Raw text	88.69%
HTML removal	88.73%
Lowercasing	88.73%
Punctuation removal	88.73%
Stop-word removal	88.00%

HTML removal produced a small improvement in accuracy. Lowercasing and punctuation removal did not change the reported accuracy at four-decimal precision.

Stop-word removal reduced accuracy from 88.69% to 88.00%. This suggests that common words can carry useful information for sentiment classification. Negation and context are particularly important when interpreting opinions.

3.2 Feature Representation Comparison

CountVectorizer and TF-IDF were compared using five-fold cross-validation.

Feature Representation	Mean Cross-Validation Accuracy
CountVectorizer	88.73%

Feature Representation	Mean Cross-Validation Accuracy
TF-IDF with unigrams	89.35%
TF-IDF with unigrams and bigrams	90.12%

TF-IDF with unigrams improved accuracy by 0.62 percentage points over raw counts.

Adding bigrams increased accuracy by a further 0.77 percentage points. Bigrams allow the model to learn short word combinations, such as *not good* or *very enjoyable*, which may convey sentiment more effectively than isolated words.

These results indicate that TF-IDF with unigram and bigram features is a suitable representation for this dataset.

4. Model Development and Hyperparameter Tuning

Four machine learning algorithms were evaluated:

1. Logistic Regression
2. Multinomial Naive Bayes
3. Linear Support Vector Machine (SVM)
4. Random Forest

Hyperparameter tuning was performed using cross-validation on the training data.

4.1 Cross-Validation Results

Model	Best Parameters	Mean CV Accuracy
Linear SVM	C = 1	91.34%
Logistic Regression	C = 10	91.21%
Multinomial Naive Bayes	alpha = 0.1	88.81%
Random Forest	max_depth = None, n_estimators = 100	84.55%

Linear SVM achieved the highest mean cross-validation accuracy at 91.34%, closely followed by Logistic Regression at 91.21%.

Random Forest used a reduced unigram feature space to manage computational cost. Consequently, its feature representation differed from that of the other three models, so its result is not a perfectly controlled comparison.

5. Final Model Evaluation

The tuned models were evaluated on a held-out test set containing 10,000 reviews.

The evaluation used the following metrics:

- **Accuracy:** Proportion of all predictions that were correct.
- **Precision:** Proportion of predicted positive reviews that were actually positive.

- **Recall:** Proportion of actual positive reviews correctly identified.
- **F1-score:** Harmonic mean of precision and recall.
- **ROC-AUC:** Measures how effectively a model distinguishes positive and negative reviews across classification thresholds.

5.1 Test Results

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	**91.45%**	91.14%	91.78%	**91.46%**	97.01%
Linear SVM	91.20%	90.74%	91.72%	91.22%	**97.03%**
Multinomial Naive Bayes	89.47%	88.97%	90.05%	89.51%	95.91%
Random Forest	84.56%	85.14%	83.64%	84.38%	92.61%

Logistic Regression achieved the highest test accuracy and F1-score, making it the selected final model.

Linear SVM achieved a marginally higher ROC-AUC of 97.03%, compared with 97.01% for Logistic Regression. Both linear models demonstrated strong classification performance.

Random Forest performed less effectively in this experiment, potentially due in part to the reduced feature representation used to control computational cost.

5.2 ROC Curve Analysis

ROC curves were generated for all four models and plotted on the same graph.

The ROC curves for Logistic Regression and Linear SVM lie close to the upper-left region of the graph, indicating strong discrimination between positive and negative reviews.

The AUC values were:

- Logistic Regression: 0.9701
- Linear SVM: 0.9703
- Multinomial Naive Bayes: 0.9591
- Random Forest: 0.9261

All models performed better than the random-classifier baseline.

5.3 Confusion Matrix

The confusion matrix for Logistic Regression showed that the model correctly classified most reviews in both classes.

The classification report gave approximately 0.91 precision, recall, and F1-score for both positive and negative reviews. This indicates balanced performance rather than strong results for only one sentiment class.

The confusion matrix is saved as results/figures/confusion_matrix.png.

6. Model Interpretability

Logistic Regression coefficients were examined to identify the text features most strongly associated with each sentiment.

Strong positive features

Examples include:

- great
- excellent
- perfect
- wonderful
- hilarious
- amazing

Strong negative features

Examples include:

- worst
- awful
- boring
- bad
- waste
- poor

Positive coefficients push the model toward predicting positive sentiment, while negative coefficients push it toward negative sentiment.

The model also learned useful short phrases, such as *the worst* and *must see*, because the TF-IDF representation includes both unigrams and bigrams.

These coefficients represent learned associations, not universal rules. A word's meaning depends on the context in which it appears.

The feature coefficient visualization is saved as results/figures/top_sentiment_features.png.

7. Learning Curve Analysis

A faster, approximate learning-curve experiment was conducted using Logistic Regression. TF-IDF was fitted once on a 12,000-review subset, and the model was trained on progressively larger subsets with a fixed validation set.

Training Samples	Training Accuracy	Validation Accuracy
2,000	100.00%	85.62%

Training Samples	Training Accuracy	Validation Accuracy
10,000	99.93%	89.31%
20,000	99.70%	90.52%

Validation accuracy increased as the training sample size grew. However, training accuracy remained close to 100%, indicating a substantial gap between training and validation performance.

This suggests overfitting, particularly with smaller training subsets. The narrowing gap indicates that additional training data may improve generalization.

Limitation: This is an approximate learning curve using a fixed validation split, not the same three-fold cross-validation procedure used in the main model comparison.

The visualization is saved as results/figures/learning_curve.png.

8. Misclassification Analysis

Thirty misclassified test reviews were sampled for manual inspection.

The sample contained:

- 13 negative reviews incorrectly predicted as positive.
- 17 positive reviews incorrectly predicted as negative.

Preliminary inspection suggested several possible causes of classification errors:

1. **Mixed sentiment:** A review may praise certain aspects while criticizing others.
2. **Context-dependent wording:** The sentiment of a phrase may depend on surrounding text.
3. **Contrastive statements:** Expressions such as “I expected the film to be bad, but I was wrong” require contextual interpretation.
4. **Subtle sentiment:** Some reviews describe a plot without using obvious positive or negative words.
5. **Sarcasm and irony:** Literal wording may not match the reviewer's intended opinion.

These observations illustrate the limitations of TF-IDF-based models, which primarily rely on word and phrase statistics rather than a deep understanding of context.

The 30 sampled errors are saved in results/misclassified_reviews.csv.

The observations are qualitative and should not be interpreted as measured frequencies for the entire test set.

9. Limitations

Despite the strong results, several limitations remain:

- TF-IDF does not fully capture semantic meaning or long-range context.
- Sarcasm, irony, and mixed sentiment can be difficult to classify.

- Random Forest used a reduced feature representation, limiting direct comparability with the other models.
- The learning-curve experiment used a subset of the data and a fixed validation split.
- Performance on this dataset may not generalize to reviews from other platforms or domains.

10. Conclusion

This project investigated sentiment classification on the IMDB dataset using preprocessing experiments, text feature engineering, and four machine learning algorithms.

The experiments showed that TF-IDF outperformed raw term counts and that adding bigrams improved cross-validation accuracy. Stop-word removal reduced performance, demonstrating that more preprocessing does not necessarily produce better results.

Among the evaluated classifiers, Linear SVM achieved the highest cross-validation accuracy of 91.34%, while Logistic Regression achieved the highest held-out test accuracy of **91.45%** and F1-score of **91.46%**.

Logistic Regression was selected as the final model because of its strong test performance and interpretable coefficients. Its ROC-AUC of 97.01% indicates strong discrimination between positive and negative reviews.

Overall, TF-IDF with unigram and bigram features combined with a linear classifier proved effective for this binary sentiment-classification task.

11. Tools and Libraries

The project used:

- Python
- Pandas
- NumPy
- scikit-learn
- Matplotlib
- Beautiful Soup
- Jupyter Notebook

12. References

1. IMDB Dataset of 50K Movie Reviews, Kaggle.
2. scikit-learn documentation.
3. Pandas documentation.
4. Matplotlib documentation.
5. Beautiful Soup documentation